Autonoid Racing: Technical Blueprint v2.0

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1. System Architecture

This blueprint defines the Autonoid 1/8 scale off-road platform. The design prioritizes proprioceptive awareness over traditional SLAM-based vision, utilizing a custom "Rhythm-Based Controller" to map physical track signatures.

2. Zero-Noise Interface: Decoupling Engineering

To ensure high-fidelity sensor data, the platform utilizes active mechanical decoupling to isolate compute and sensors from chassis vibration.

- **Suspension-Integrated Sensor Arms:** Sensors are mounted in hollowed, 3D-printed suspension arms with Shore A 20-30 silicone sleeves for high-frequency damping.
- **Under-Chassis Optical Array:** A "Dark Tunnel" housing (15mm depth) creates an air-damped aperture, protecting the optical flow sensor from debris and ride-height shadows.
- **Sandwich Compute Platform:** The NVIDIA Jetson compute stack is isolated via a mass-damper-spring mount utilizing alpha-gel dampers to create a mechanical low-pass filter.

3. Performance Metrics (Autonoid Grades)

Metric	Focus	Target Grade
PS-Grade	Proprioceptive Symmetry	A
RA-Grade	Rhythm Adherence	A-
DA-Grade	Dynamic Adaptability	B+

4. Hardware Bill of Materials (Core)

- **Chassis:** 1/8 Scale 4WD Electric Buggy (e.g., ARRMA platform).
- **Compute:** NVIDIA Jetson Orin Nano.
- **MCU:** Teensy 4.1 for low-latency sensor polling.
- **Damping:** Kyosho Zeal Gel + Silicone Alpha-gel mounts.

5. Implementation Blueprint

1. **Structural Foundation:** Integration of drive-by-wire and vibration-isolated sensor mounting.

2. **Feedback Integration:** Calibrating proprioceptive hub-IMU data with optical flow ground-truth.
3. **Cognitive Deployment:** Training the "Corner Signature" rhythm engine in simulated race environments.